

Attention in deep learning is a mechanism that allows models to dynamically focus on the most relevant parts of an input sequence by assigning varying importance weights to different elements.

Why Attention Was Created

Traditional models like Recurrent Neural Networks (RNNs) and LSTMs process text or sequential data step-by-step. They compress entire inputs into a single fixed-size vector. This creates an information bottleneck, causing the model to forget earlier words in long sentences.

The attention mechanism was introduced by Bahdanau et al. in 2014 to let the decoder look back at all encoder hidden states instead of relying on a single compressed vector.

How Self-Attention Works in Transformers

Introduced in the 2017 landmark paper "*Attention Is All You Need*", the [Transformer](#) architecture uses **self-attention** to replace recurrence entirely. It computes relationships between every token in a sequence simultaneously using three learnable matrices:

- **Queries (Q):** What information a token is looking for.
- **Keys (K):** What information a token holds to match against queries.
- **Values (V):** The actual content extracted and passed forward once a match is found.

Scores are calculated via dot products, scaled, and normalized using a softmax function to create an attention weight distribution.

Main Benefits of Attention

- **Solves Long-Range Dependencies:** Direct paths between any two words mean distance no longer dilutes context.
- **Enables Parallelization:** Unlike RNNs that must wait for previous steps to finish, attention computes all token interactions at the same time, speeding up GPU training.
- **Improves Interpretability:** Attention weights show which input tokens most influenced a model's prediction